



Study of optical properties of GeO₂ nanocrystals as synthesized by hydrothermal technique

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ABSTRACT

The GeO₂ nanocrystals (α -quartz type structure) with β -phase are synthesized at relatively lower temperature by hydrothermal route using autoclave. All samples are characterized by XRD, FESEM, EDS, TEM, photoluminescence (PL) and UV–vis absorption spectroscopy techniques. Synthesized nanocrystals have uniform shape and uniform size distribution for a particular synthesis condition, which is about 30–300 nm depending on synthesis conditions. The XRD results indicate that grown GeO₂ crystals only shows peak related to α -quartz structure with lattice parameters $a = 4.985$ Å and $c = 5.648$ Å. UV–vis absorption spectroscopy measurements reveal the bandgap energies corresponding to the GeO₂ α -quartz structure. Synthesized nanocrystals are capable to emit strong blue light around 425–435 nm under excitation of 300 nm and 325 nm and consequently the as synthesized material can be used in integrated optical devices.

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1. Introduction

GeO₂ nanomaterial is a blue photo luminescent material being of great deal of interest due to their unique and fascinating physical and chemical properties. GeO₂, a transparent conductive oxide (TCO), is used as an important optical material for optical fiber and waveguide [1,2]. At the same time, it has great potential to be used as nanoconnector in optical or optoelectronic nano or micro devices [3,4]. Luminescent GeO₂ exhibits high value of dielectric constant, refractive index, thermal stability and mechanical strength [4]. GeO₂ based glass shows higher refractivity and linear thermal expansion coefficient than the corresponding of SiO₂ [3–5]. Due to these attractive optical and electronic properties, presently the GeO₂ based nanomaterial is considered as one of the promising materials for different applications including optoelectronic devices, photosensors, vacuum technology, etc. [6–14]. Several methods of synthesis of GeO₂ nanomaterial are reported till now. GeO₂ nanowires and nanorods have been synthesized by thermal treatment of Ge [2,3], carbothermal reaction [6], carbon nanotube confined reaction of Ge [15], electro-spinning [16], laser ablation [17], etc. GeO₂ nanoparticles have been prepared by DC magnetron sputtering and heat annealing, solution based method. Chemical precipitation [4,18] and thermal evaporation [5,8,19] are widely used techniques to synthesize nanocrystals of GeO₂.

Synthesis techniques need optimization to control shape, size as well as development of particular phase of crystals. GeO₂ shows several polymorphs including tetragonal α phase with rutile type structure and tetragonal/rhombohedral β phase with α -quartz structure [4]. This TCO has higher bandgap energy (direct bandgap energy > 5 eV) than any other TCO, which enables it to act as host material for optical impurities aiming towards the development of any luminescent device in the range of UV–blue to NIR [20]. It is studied as an alternative host material for plane waveguides due to its low phonon energies and hence improved quantum efficiency. Like other semiconductor materials GeO₂ also has oxygen related defects. Mostly these defects play an active role for their smart optical and electronic properties, which can be used as a key factor to design optical devices.

In our present study, β -phase GeO₂ nanocrystals with α -quartz type structure are synthesized by hydrothermal method in different pressure–temperature conditions. Surface morphology, structure and size distribution of synthesized nanocrystals are systematically studied by using FESEM, TEM and XRD. The present work also shows the optical absorption properties and photoluminescence properties of the samples.

2. Experimental

Analytical grade GeO₂ powder (Alfa Aesar, 99.99%) is used as starting materials. At first a stock solution of GeO₂ with sparingly hot water is prepared. Then a required amount of the solution is taken in the in-house designed reaction kettle (moc: SS-316, maximum designed pressure 200 bar, volume: 500 cm³), attached

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Table 1Temperature and pressure of the synthesis of GeO₂ nanocrystals.

Sample no.	Temperature (°C)	Pressure (bar)
Ge1	100	1
Ge2	185	11
Ge3	215	21
Ge4	285	69

with a pressure gauge. The kettle is then properly sealed and placed in a furnace for heating in different temperatures for 24 h. The pressures of the synthesis at different time have been recorded from the pressure gauge attached to kettle. The reaction kettle is then naturally cooled to ambient temperature and the samples are collected from the product solution by the method of evaporation of water. The temperature and pressure of the synthesis are tabulated in Table 1. Structural properties of the all samples are studied by X-ray diffraction (XRD) analysis. XRD patterns are recorded by Rigaku MiniFlex2 instrument with Cu K α radiation. Microstructure and surface morphology of all samples are studied by FESEM using Hitachi S-4800. Energy dispersive X-ray spectroscopy (EDS) and scanning electron microscopy (SEM) of the samples are performed by Hitachi S3400N instrument. An idea about the crystal shape and size distribution is estimated by TEM (TECNEI G2). By using Spectrofluorometer (Horiba Jobin Yvon, Model: SPEX Fluoromax 4P, Source: 150 W ozone-free xenon arc lamp) PL emission spectrum of the all samples are obtained in room temperature. The excitation wavelengths were 300 nm

and 325 nm. At the same time UV–vis spectrophotometer (Perkin-Elmer, Model: Lambda 45) is used for absorption spectroscopy study.

3. Result and discussion

Fig. 1(a) and (b) shows the XRD histogram of the samples Ge3 and Ge4, respectively, which confirm good crystalline character of the materials. All the peaks are in well agreement with those of: β -phase GeO₂ crystals with α -quartz type structure with the lattice parameters $a = 4.985 \text{ \AA}$ and $c = 5.648 \text{ \AA}$ as mentioned in PDF-43-1016. All recognizable peaks including (1 0 0), (1 0 1), (1 1 0), (1 0 2), (1 1 1), (2 0 0), (1 1 2), (2 0 2), (2 1 1), (1 1 3), (2 0 3), (0 1 4) and (3 0 2) are clearly indexed to the hexagonal GeO₂. It can be inferred from XRD results that higher temperature of synthesis as well as pressure of synthesis do not cause any phase change. Fig. 1(c) shows the XRD spectrum of the starting material. The SEM image of starting material (analytical grade GeO₂ powder) in Fig. 2(a) shows bulk GeO₂ crystals. Fig. 2(b)–(d) shows FESEM images of GeO₂ nanocrystals synthesized at 100 °C, 185 °C and 285 °C, respectively. Surface morphology and distribution analysis of the samples in Fig. 2(b) indicate that the sizes of the GeO₂ particles are ranging from 100 nm to 300 nm and particles are agglomerated and forming network to become stack shaped. Fig. 2(c) shows that the crystals tend to lose agglomeration as the pressure and temperature of synthesis increased. In this case a synthesis temperature 185 °C and pressure 11 bar might be helping crystals to lose their integrity by providing sufficient

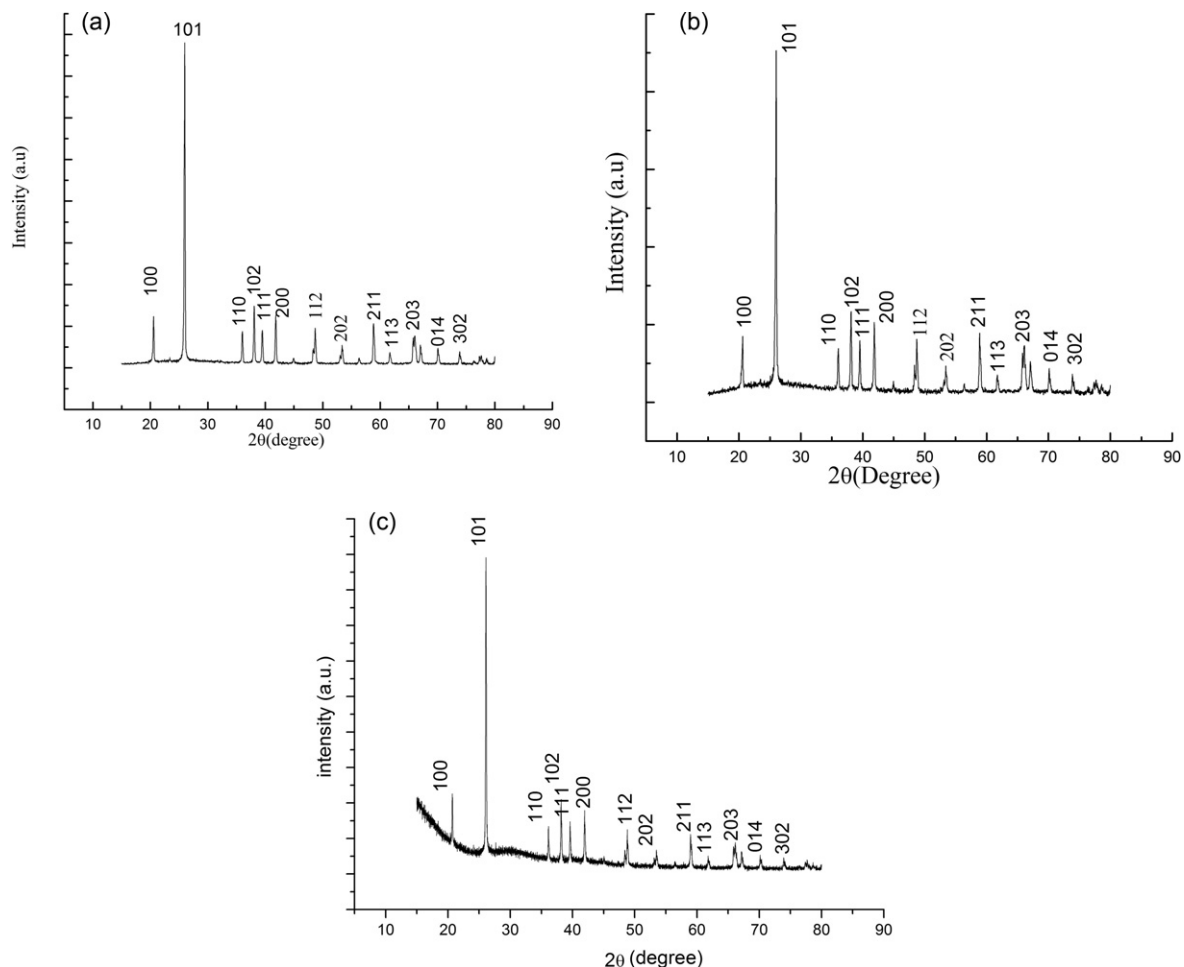


Fig. 1. XRD histogram of (a) GeO₂ samples synthesized at 215 °C, (b) GeO₂ samples synthesized at 285 °C and (c) starting material (GeO₂ powder).

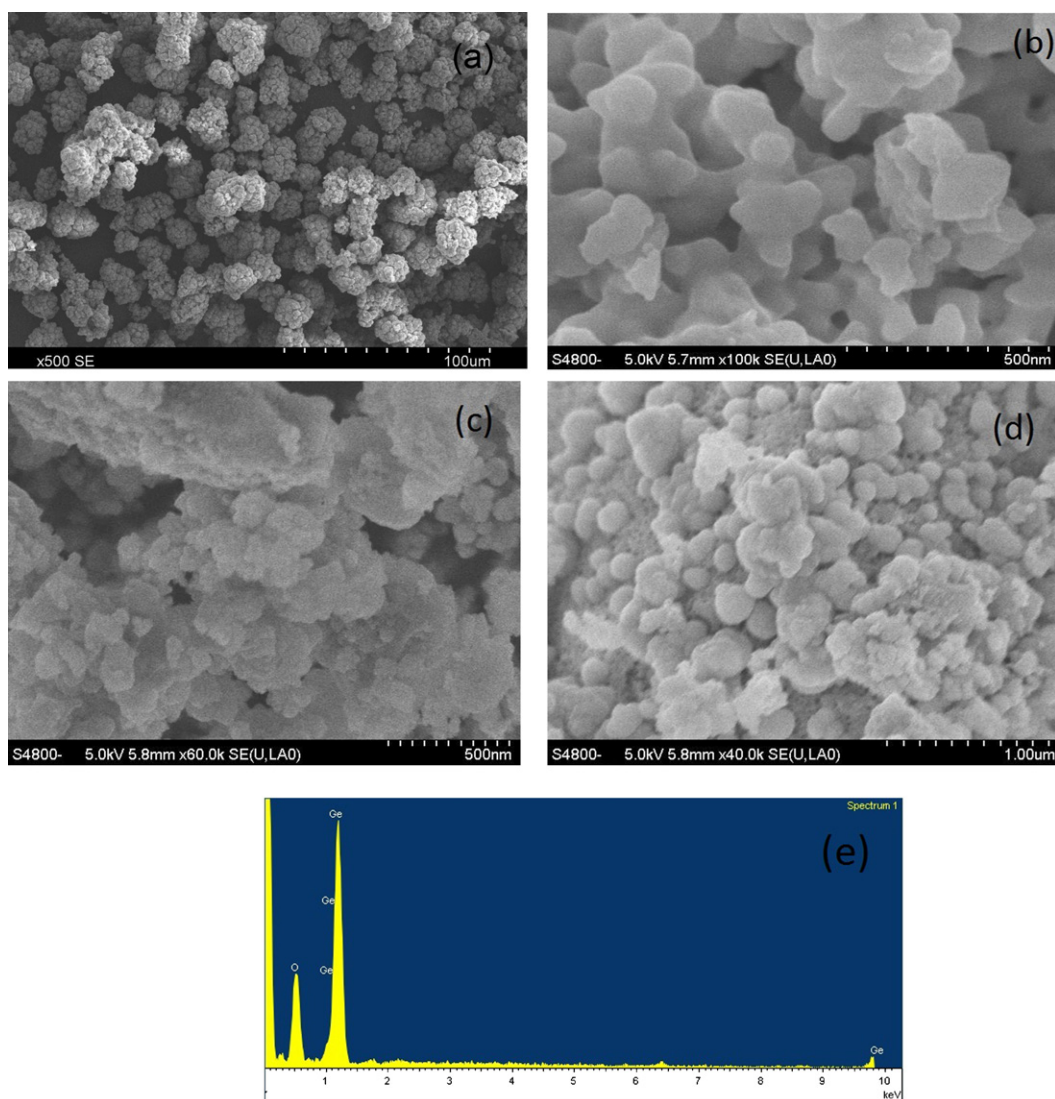


Fig. 2. (a) SEM image of starting material (GeO₂ powder); FESEM image of GeO₂ nanocrystals synthesized at (b) 100 °C, (c) 185 °C and (d) 285 °C; (e) EDS result of the sample synthesized at 185 °C.

energy. At the synthesis temperature 285 °C samples change their morphology to a greater extent (Fig. 2(d)). The grains are well separated from stack like agglomeration and they now start to agglomerate into a new network of rounded shape and flower petals type architecture. It is evident from these figures that the surface morphology changes with the increasing pressure (and temperature) of synthesis. The grains of the samples get separated apart as the pressure of synthesis increased and at last become individual round shape grain. The required energy for the formation of individual grain is supplied due to the increment of pressure and temperature of synthesis of the particles. On the other hand EDS results (Fig. 2(e)) confirm about the constituents of the samples, which are only Ge and O since the characteristic peaks of Ge and O only appear in EDS spectra. The TEM micrographs of some samples are shown in Fig. 3. It is evident from the images that nanocrystals are properly formed, whose size distribution is uniform for a particular pressure (or temperature) of the synthesis. TEM micrograph of the sample Ge1 in Fig. 3(a) shows the synthesized hexagonal GeO₂ nanocrystal with size ranging 75–300 nm. Though the size distribution is not quite uniform in this synthesis condition but the hexagonal shape is common to all nanocrystals. Fig. 3(b) shows the TEM image of the sample Ge3 synthesized at 215 °C. The image indicates that the sizes of

synthesized hexagonal, round shaped nanocrystals are reduced and these are ranging from 40 nm to 60 nm. The size distribution is very uniform. TEM image of the sample Ge4 in Fig. 3(c) confirms that the hexagonal, round shaped nanocrystals with uniform size distribution (size: 30–50 nm) have been formed. All the XRD, EDS, FESEM and TEM results ensure that the synthesized GeO₂ nanocrystals exhibit high structural quality in terms of physical and chemical structure. It should be mentioned here that we have synthesized β -phase α -quartz type GeO₂ nanocrystals at comparatively lower temperature, which are smaller in size in comparison to that of other lower temperature synthesis work [4,18]. It is also to be noted that as the pressure of synthesis increases the size of the particles diminishes. So based on our study further attempts may be performed to optimize the pressure of synthesis to get particular size distribution of nanocrystals.

The optical absorption curve for the GeO₂ nanocrystal sample synthesized at 215 °C (sample: Ge3) is shown in the Fig. 4(a). The enlarged view of the selected area is shown in the inset of the figure. Fig. 4(a) shows that absorption peaks occur around at 200 nm, 216 nm, 233 nm, 245 nm and 335 nm. For all other synthesized samples these five absorption peaks also occur at almost same positions. From this optical measurement it can be inferred about the electronic transitions, which occur at different

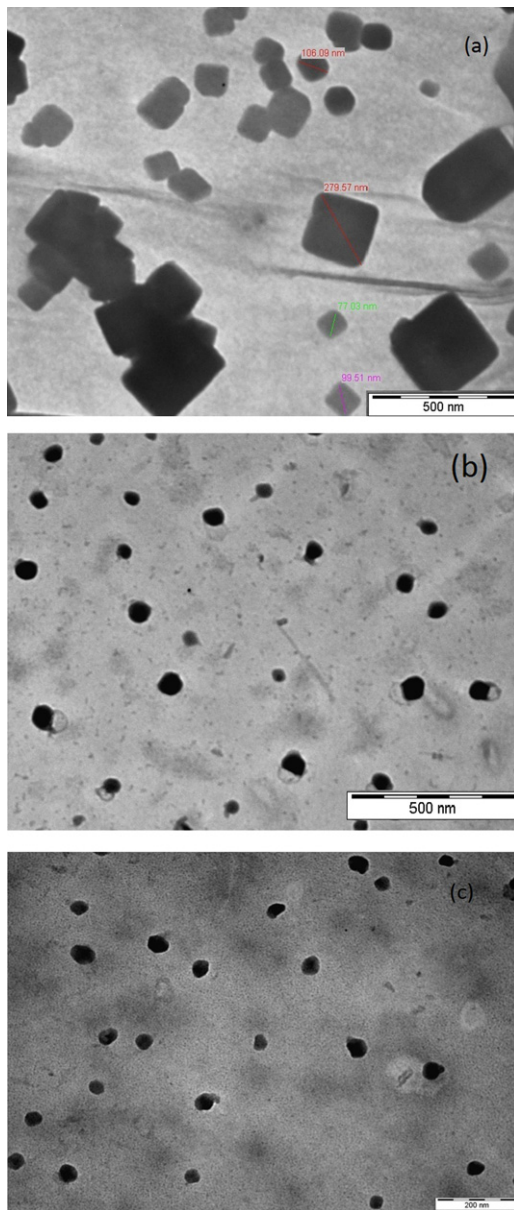


Fig. 3. TEM image of sample synthesized at (a) 100 °C, (b) 215 °C and (c) 285 °C.

valence to conduction band as well as the response due to oxygen vacancies. Recently Liu et al. [21] has published an exhaustive theoretical work for band structure calculation of α -quartz GeO_2 crystals using plane wave ultrasoft pseudopotential techniques, based on the first principles density function theory (DFT). But to the best of our knowledge no experimental finding supporting the work of Liu et al. has been reported till now. Our results of optical absorption spectroscopy are in fair agreement with the theoretical findings obtained by Liu et al. The absorption spectrum shows the peaks at 200 nm and 245 nm, which are due to oxygen vacancies in the sample [22], whereas the other peaks confirm the transition of electrons within the band gap of α -quartz GeO_2 crystal as calculated by Liu et al. Among these the peak at 216 nm (corresponding energy 5.74 eV, $\text{K} \rightarrow \text{K}$ valence to conduction band transition) is also in good agreement with the findings of Ramana et al. [4]. But the peaks around 233 nm (energy 5.32 eV, $\text{K} \rightarrow \text{H}$ valence to conduction band transition), 270 nm (4.6 eV) and 335 nm (energy 3.7 eV, $\Gamma \rightarrow \Gamma$ valence to conduction band transition) are not reported by them. All these peaks, which occur

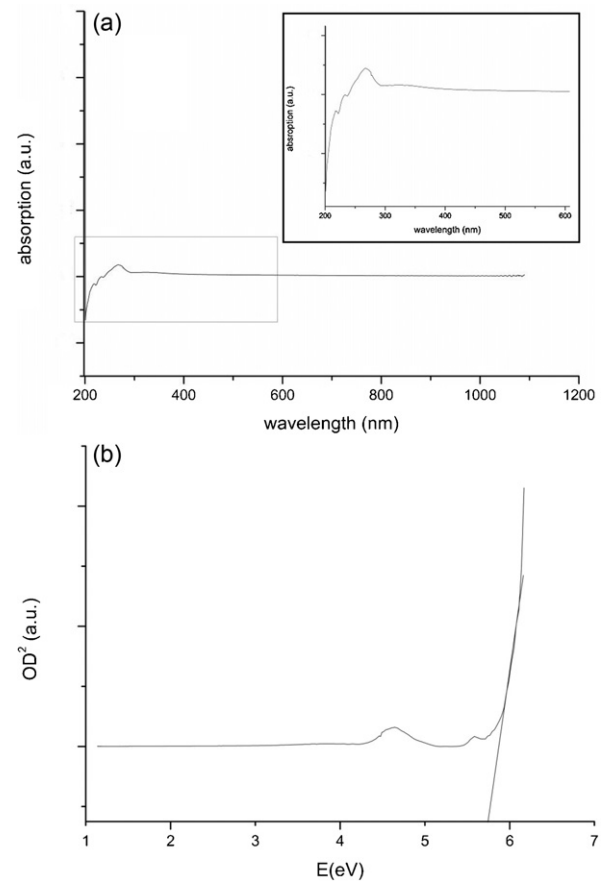


Fig. 4. (a) UV-vis absorption spectroscopy results of GeO_2 nanocrystals synthesized at 215 °C. Inset shows enlarged view of the selected area. (b) Plot of $(\text{OD})^2$ vs. incident photon energy for GeO_2 nanocrystals synthesized at 215 °C.

for as synthesized samples, support the calculated band gap energy for bulk α -quartz GeO_2 crystals [21]. Thus it can be concluded that as synthesized nanocrystals also have similar bandstructure like bulk materials. In Fig. 4(b) square of the optical density (OD) is plotted against the incident photon energy from which the direct band gap of the sample Ge3 is evaluated according to [4]. The value of the direct bandgap is evaluated as 5.74 eV, which is very much similar with the same of [4].

Fig. 5(a) and (b) shows the PL emission spectrum of the sample Ge3 and Ge4, respectively under two different photoexcitations 300 nm and 325 nm at room temperature. It is clearly observed that all samples emit strong blue lights under excitation at 300 nm and 325 nm. Fig. 5(a) indicates that the sample Ge3, i.e. GeO_2 nanocrystals synthesized at 215 °C show broad blue emission bands centered around 427 nm and 430 nm under 300 nm and 325 nm excitation, respectively. Fig. 5(b) indicates that sample Ge4 shows blue emission around at 428 nm and 434 nm under 300 nm and 325 nm excitations, respectively. Sample Ge2 also shows emission around 425–435 nm (not shown here). For excitation of 300 nm and 325 nm every sample shows small peaks at 335 nm and 365 nm, respectively, which may cause due to Raman scattering. This is confirmed by taking the PL emission spectra of the samples at different excitation energies. Fig. 5(c) shows the PL emission spectrum of the starting material (GeO_2 powder) under 300 nm and 325 nm excitation. It is evident from the figure that starting material is not showing any significant photoluminescence property as expected. Only sharp peaks associated with scattering phenomenon are prominent. No intense emission peak other than scattering related peak is observed in the figure. So the

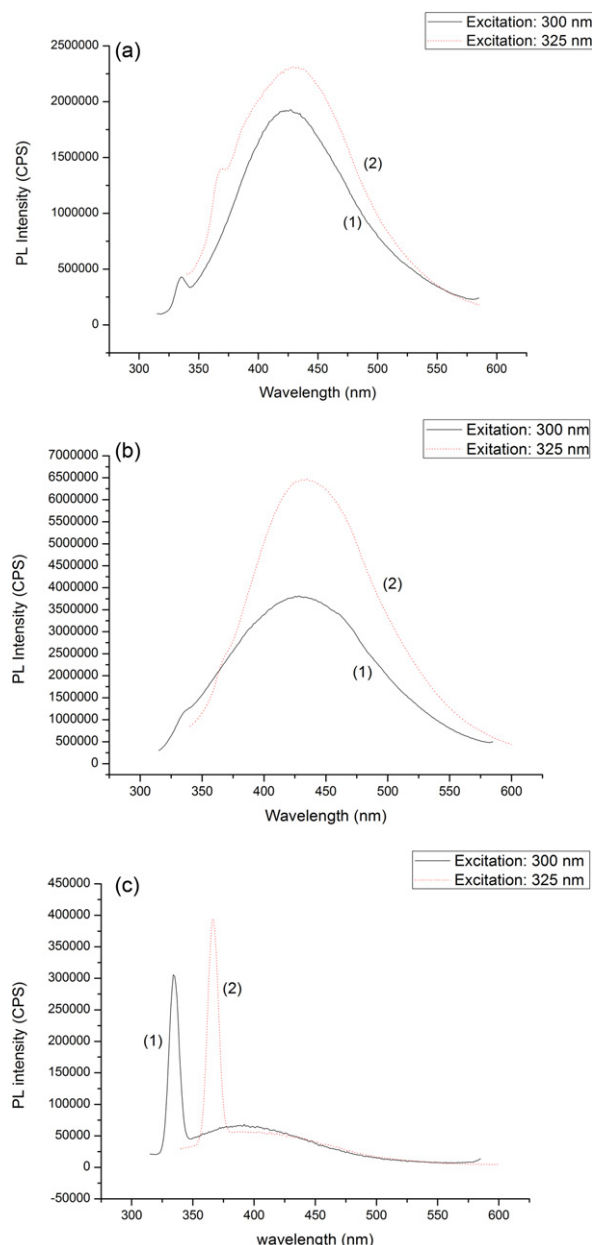


Fig. 5. PL emission spectra of the (a) samples synthesized at 215 °C, under excitation at: (1) 300 nm and (2) 325 nm; (b) samples synthesized at 285 °C, under excitation at: (1) 300 nm and (2) 325 nm; (c) starting material (GeO_2 powder) under excitation at: (1) 300 nm and (2) 325 nm.

synthesized nanocrystals are showing different and much better photoluminescence characteristics than the starting material. Zacharias et al. [23] have studied light emission from Ge and GeO_2 nanocrystals embedded in SiO_x matrix. They have reported blue luminescence near 400 nm wavelength. This PL peak position for our work is quite different from the findings presented in that work as the crystal sizes are very much different from the present work. In general commercial GeO_2 powder gives peak around 536 nm [24]. Moreover PL emission from spectrally pure GeO_2 powder is found to be less intense [6]. Blue and green light emission has been previously observed from GeO_2 nanowires, nanotubes and nanofibers [3,6,8,20,24]. Corresponding peak positions are around 427 nm, 507 nm, 485 nm, 448 nm, 516 nm and 563 nm, respectively. Blue and green light emissions are originated from radiative recombination corresponding to defects

in GeO_2 crystals, including oxygen vacancy and oxygen–germanium vacancy center [3–6]. Though in our work the GeO_2 nanocrystals are produced at comparatively lower temperature but still large number of vacancies are present in synthesized β -phase α -quartz GeO_2 nanocrystals, which can be attributed for blue light emission. In the present work introduction of large amount of vacancies at lower synthesis temperature is also evident from absorption spectroscopy. Several works have been previously reported [2,3,5,6] regarding the vacancies in α -quartz GeO_2 crystal structure, but synthesis temperatures ($\sim 900^\circ\text{C}$) were much higher for those works in comparison to our work. In our present work we achieve significant photoluminescent properties of as synthesized GeO_2 nanocrystals, which was not previously reported in the works [4,18], in which the β -phase GeO_2 nanocrystals were synthesized with α -quartz type structure at lower temperature.

4. Conclusion

To the best of our knowledge for the first time, β -phase GeO_2 α -quartz nanocrystals with average size less than 100 nm are synthesized by using simple hydrothermal method at sufficiently lower temperature (as low as 100°C), which are traditionally formed at higher temperature ($\sim 1000^\circ\text{C}$) in general. XRD, microscopic characterization of the samples reveal that synthesized nanocrystals exhibit uniform shape and size distribution as well as high structural quality and phase purity. The as synthesized nanocrystals have size distribution from 30 nm to 300 nm depending on the synthesis condition. The PL emission spectrum of GeO_2 nanocrystals shows blue emission band around 425–435 nm, which is caused due to oxygen and oxygen–germanium related vacancies. Existence of oxygen deficiency in crystal structure is also confirmed from UV–vis absorption spectroscopy results. These vacancies are introduced in these types of GeO_2 nanocrystals at comparatively lower temperature for the first time by hydrothermal method. UV–vis absorption spectroscopy results are in good agreement with theoretically calculated bandgap energies for α -quartz GeO_2 crystals. Absorption spectroscopy of the samples also reveals that the band structure for GeO_2 nanocrystals is similar to that of the bulk crystals. From the results of our studies it can be concluded that as synthesized nano-sized GeO_2 nanocrystals with fascinating optical properties can be successfully used for fabrication of nanophotonic devices.

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